Determine the Area of Triangles

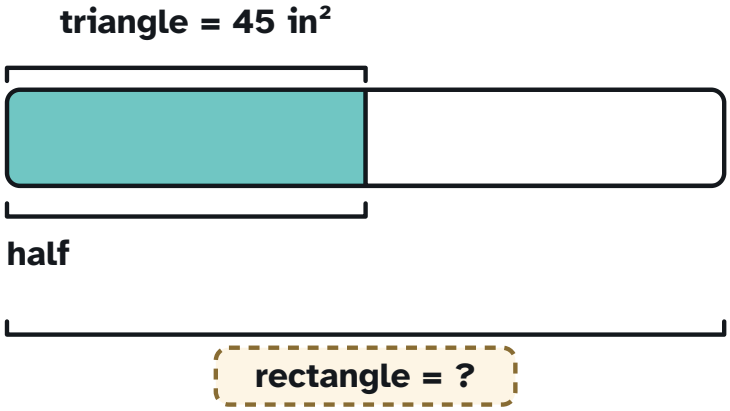
Find a missing base or height of a triangle when you know its area.

Name _____ **Date** _____

Tonight's spark

A triangular flag has a tag that says 45 square inches. The flag is 9 inches tall. How long is its base?

Make a guess first: is the base longer or shorter than the height?



Our class model: work the formula backwards

Try it

Work it out on paper. Write one answer on each line.

1 **To undo the ½ in A = ½ × b × h, what do you do to the area first?**

A. Halve it **B.** Subtract 2 **C.** Double it **D.** Add the base

My answer _____

2 **A triangle has an area of 12 cm² and a base of 6 cm. What is its height?**

A. 4 cm **B.** 2 cm **C.** 72 cm **D.** 18 cm

My answer _____

3 **A triangle has an area of 30 m² and a height of 5 m. What is its base?**

A. 6 m **B.** 150 m **C.** 35 m **D.** 12 m

My answer _____

4 **A triangle has an area of 18 and a base of 9. Which equation can you use to find its height h?**

A. ½ × 9 × h = 18 **B.** 9 × h = 18 **C.** ½ × 18 × h = 9 **D.** 9 + h = 18

My answer _____

5 **A triangle has an area of 28 cm² and a base of 7 cm. What is its height?**

A. 4 cm **B.** 21 cm **C.** 8 cm **D.** 196 cm

My answer _____

Talk about it

The area of a triangle is 12 and its base is 8. What do you do first to undo the half?

Home mission

Say, "A flag has an area of 45 square inches and a height of 9 inches." Ask your student to explain how to find the base. What do you do first? Then ask them to check the answer in the formula.

If we get stuck, ask

- What does the formula say, and which part is missing?
- What undoes taking half of something?
- Put your answer back in. Does the area come out right?

Do it on a screen

Scan or type the link. Same lesson, with hints.

eduwonderlab.com/lessons/5-2-part2/homework.html

How did it feel? (optional)

Tricky Getting there Got it